

Discrete Random Variables & Probability Distributions

STAT 521

Overview

- ▶ Discrete vs continuous random variables
- ▶ Probability mass function (pmf) and cumulative distribution function (cdf)
- ▶ Expectations and variance
- ▶ Common discrete distributions: uniform, Bernoulli, binomial, geometric, negative binomial, hypergeometric, Poisson

Random Variables

A **random variable** is a function that assigns a real number to each outcome of a random experiment: $X : S \rightarrow \mathbb{R}$.

Capital letters denote variables (e.g., X, Y) and lowercase denote realizations (e.g., x, y).

- ▶ **Discrete:** range is finite or countably infinite (e.g., counts).
- ▶ **Continuous:** range is an interval on $\mathbb{R}$ (e.g., lengths, times).

Example: Discrete random variables

For each outcome $\omega \in S$, a random variable X assigns a real number $X(\omega)$.

Example: Die Roll sample space:

$$S = \{1, 2, 3, 4, 5, 6\}.$$

Define

$$X(\omega) = \omega \quad (\text{identity mapping})$$

Thus X can take values $x \in \{1, 2, 3, 4, 5, 6\}$, which is the **support** of X . X is a discrete random variable.

Example: Continuous random variable

A piece of arbitrary length is cut from a cable of length 25 inches.

Let X be the length of the cut piece. Then,

$$\{x \mid 0 < x < 25\}$$

X is a continuous variable.

Probability Mass Function (pmf)

- ▶ The probability distribution of X describes the probabilities associated with the possible values of X .
- ▶ For discrete X , this is a list of values given by a **probability mass function** (pmf).
- ▶ Equivalently, it is a collection of probabilities attached to discrete points on the number line.

Probability Mass Function (pmf)

Definition. Let X be a discrete random variable with possible values $\{x_1, x_2, \dots, x_n, \dots\}$. A function

$$f_X(x) = \mathbb{P}(X = x)$$

is a probability mass function if it satisfies

1. $f_X(x_j) \geq 0$ for all j ,
2. $\sum_j f_X(x_j) = 1$.

$f_X(x_j)$ denotes the probability that the random variable X has the value x_j . Outside the support $\{x_j\}$, we take $f_X(x) = 0$.

Example: Three Coins

Toss three fair coins and let X be the number of heads. The sample space is,

$$S = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}.$$

Then, $x \in \{0, 1, 2, 3\}$ and

x	0	1	2	3
$f(x)$	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$

All values of $f(x)$ are non-negative and they add up to 1.

Example

Let X take values $x = \{0, 1, 2, 3, 4\}$. Does the table below represent a probability mass function?

x	0	1	2	3	4
$f(x)$	$\frac{1}{16}$	$\frac{4}{16}$	$\frac{6}{16}$	$\frac{4}{16}$	$\frac{1}{16}$

Yes, because all values of $f(x)$ are non-negative and they add up to 1.

What is the $\mathbb{P}(X \leq 2)$?

$$\mathbb{P}(X \leq 2) = \frac{1}{16} + \frac{4}{16} + \frac{6}{16} = \frac{11}{16}.$$

Example: Toss Until First Tail

Toss a fair coin until first tail appears and let X be the number of tosses. In this case, the sample space is

$$S = \{T, HT, HHT, HHHT, \dots\}.$$

Then $x = 1, 2, 3, 4, \dots$ and the respective probabilities are

x	1	2	3	$\dots$	n	$\dots$
$f(x)$	$\frac{1}{2}$	$\frac{1}{2^2}$	$\frac{1}{2^3}$	$\dots$	$\frac{1}{2^n}$	$\dots$

Example cont.: Toss Until First Tail

Thus, $f(x) = \left(\frac{1}{2}\right)^x$, for $x = 1, 2, \dots$

To show that the values of $f(x)$ add up to 1, we must show:

$$\sum_{x=1}^{\infty} \left(\frac{1}{2}\right)^x = 1.$$

This is what we called **geometric series** in calculus. This is a probability mass function.

Cumulative Distribution Function (cdf)

For discrete X with pmf f , the cumulative distribution function is

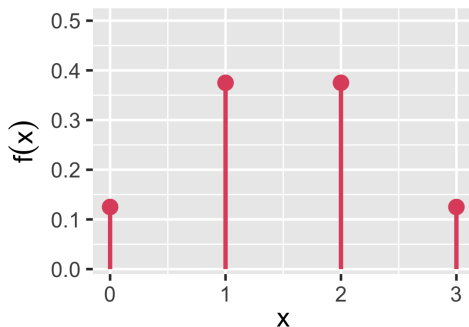
$$F_X(x) = \mathbb{P}(X \leq x) = \sum_{x_j \leq x} f_X(x_j).$$

Properties: - $F(x)$ is non-decreasing,

- ▶ $\lim_{x \rightarrow -\infty} F(x) = 0$,
- ▶ $\lim_{x \rightarrow \infty} F(x) = 1$
- ▶ $F(x)$ is bounded between 0 and 1, $0 \leq F(x) \leq 1$,
- ▶ $F(x)$ is defined for all real numbers.

Example PMF: Three Coins, number of heads

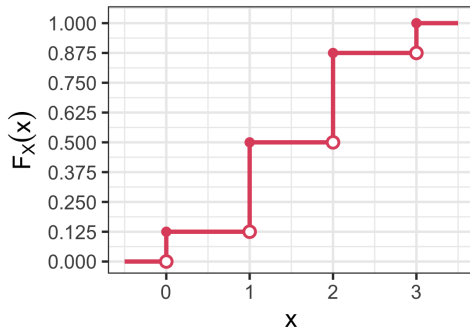
x	0	1	2	3
$f(x)$	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$



Example CDF: Three Coins, number of heads

The cumulative distribution function is the following step function:

$$F_X(x) = \begin{cases} 0, & x < 0, \\ \frac{1}{8}, & 0 \leq x < 1, \\ \frac{4}{8}, & 1 \leq x < 2, \\ \frac{7}{8}, & 2 \leq x < 3, \\ 1, & x \geq 3. \end{cases}$$



Example: PMF from CDF

- ▶ For any integer k in the support,

$$f(k) = \mathbb{P}(X = k) = F(k) - F(k^-),$$

where $F(k^-) = \Pr(X < k)$ is the left limit.

- ▶ Because cdfs are right-continuous,

$$F(k) - F(k^-) = \Pr(X \leq k) - \Pr(X < k) = \Pr(X = k),$$

the probability mass (jump size) at k .

- ▶ Graphically, $f(k)$ is the *height of the vertical jump* of the step cdf F at k .

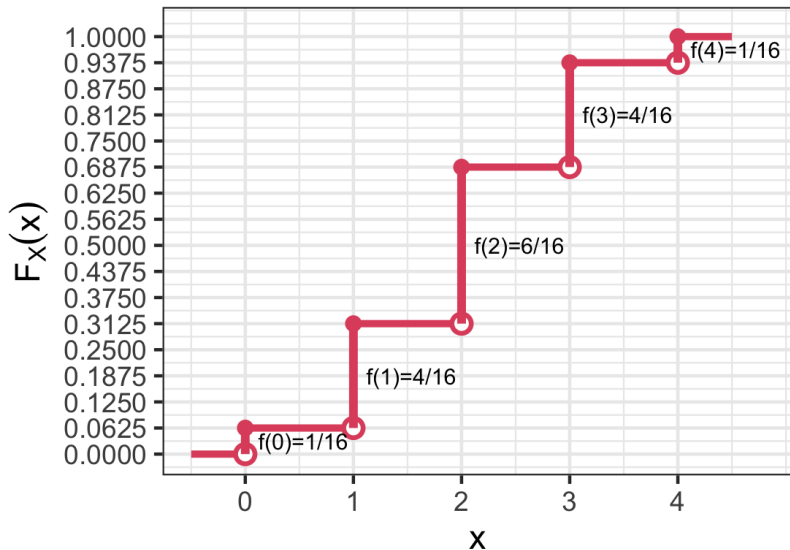
Example: PMF from CDF

Given the following CDF, find the corresponding PMF:

$$F_X(x) = \begin{cases} 0, & x < 0, \\ \frac{1}{16}, & 0 \leq x < 1, \\ \frac{5}{16}, & 1 \leq x < 2, \\ \frac{11}{16}, & 2 \leq x < 3, \\ \frac{15}{16}, & 3 \leq x < 4, \\ 1, & x \geq 4. \end{cases}$$

- ▶ $f(0) = \frac{1}{16} - 0 = \frac{1}{16}$, $f(1) = \frac{5}{16} - \frac{1}{16} = \frac{4}{16}$
- ▶ $f(2) = \frac{11}{16} - \frac{5}{16} = \frac{6}{16}$, $f(3) = \frac{15}{16} - \frac{11}{16} = \frac{4}{16}$
- ▶ $f(4) = 1 - \frac{15}{16} = \frac{1}{16}$.

Example: PMF from CDF



Expected Value

Let X be a discrete random variable with possible values $x_1, x_2, \dots$ and probability mass function $f(x)$.

Then, the **expected value** (also called the mean) of X , denoted as $\mathbb{E}[X]$ or μ , is defined as

$$\mathbb{E}[X] = \mu = \sum_{x_j} x_j f(x_j).$$

Example: Three coins, number of heads

Toss three fair coins and let X be the number of heads. The expected number of heads is:

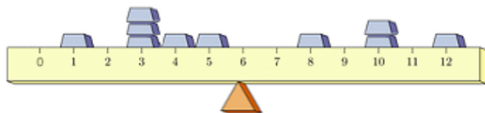
$$\mathbb{E}[X] = 0 \cdot \frac{1}{8} + 1 \cdot \frac{3}{8} + 2 \cdot \frac{3}{8} + 3 \cdot \frac{1}{8} = \frac{12}{8} = \frac{3}{2}.$$

- ▶ It doesn't mean you can get "1.5 heads" on any single toss. It's a long-run average across repetitions.
- ▶ On average half of the 3 coins are heads, so across N trials you'd expect about $1.5 \cdot N$ heads in total e.g., ≈ 1500 heads over 1000 triple-tosses.

Expected Value: Center of mass

Let X take values $\{1, 3, 4, 5, 8, 10, 12\}$ with probabilities $\{0.1, 0.3, 0.1, 0.1, 0.1, 0.2, 0.1\}$. Then,

$$\mathbb{E}[X] = 0.1 + 0.9 + 0.4 + 0.5 + 0.8 + 2.0 + 1.2 = 5.9.$$



Think of $\mathbb{E}[X]$ as the "center of mass" of the distribution.

Variance and Standard Deviation

Let X be a discrete random variable with $\mu = \mathbb{E}[X]$, then the **variance** is:

$$V(X) = \sigma^2 = \sum_{x_j} (x_j - \mu)^2 f(x_j).$$

A computationally convenient form is:

$$V(X) = \sum_{x_j} x_j^2 f(x_j) - \mu^2 = \mathbb{E}[X^2] - (\mathbb{E}[X])^2.$$

The square root of the variance, denoted as σ , is called the **standard deviation**:

$$\sigma = \sqrt{V(X)}$$

Example: Three coins, number of tails

For the three-coin pmf with $\mu = \frac{3}{2}$,

$$\begin{aligned}\sum_x x^2 f(x) &= 0^2 \cdot \frac{1}{8} + 1^2 \cdot \frac{3}{8} + 2^2 \cdot \frac{3}{8} + 3^2 \cdot \frac{1}{8} \\ &= \frac{0}{8} + \frac{3}{8} + \frac{12}{8} + \frac{9}{8} \\ &= \frac{24}{8} = 3.\end{aligned}$$

$$\begin{aligned}V(X) &= \mathbb{E}[X^2] - (\mathbb{E}[X])^2 \\ &= 3 - \left(\frac{3}{2}\right)^2 = 3 - \frac{9}{4} = \frac{3}{4}.\end{aligned}$$

Variance: Another Example

For the $\{1, 3, 4, 5, 8, 10, 12\}$ example with probabilities $\{0.1, 0.3, 0.1, 0.1, 0.1, 0.2, 0.1\}$ and $\mu = 5.9$,

$$\sum_x x^2 f(x) = 0.1 + 2.7 + 1.6 + 2.5 + 6.4 + 20 + 14.4 = 47.7,$$

$$V(X) = 47.7 - 5.9^2 = 12.89, \quad \sigma = \sqrt{12.89} \approx 3.59.$$

Special Discrete Distributions

We review:

- ▶ Discrete Uniform
- ▶ Bernoulli and Binomial
- ▶ Geometric and Negative Binomial
- ▶ Hypergeometric
- ▶ Poisson

Discrete Uniform

Suppose X can take each of the n integers $a, a + 1, \dots, b$ with equal probability, where $n = b - a + 1$. X follows a **Uniform distribution**.

Since there are n equally likely values,

$$f(x) = \Pr(X = x) = \frac{1}{n} = \frac{1}{b - a + 1}, \quad x \in \{a, a+1, \dots, b\}.$$

Discrete Uniform: Expectation

The expectation is the average of all possible values:

$$\mathbb{E}[X] = \sum_{x=a}^b x \frac{1}{n} = \frac{1}{n} \sum_{x=a}^b x.$$

The sum of consecutive integers is

$$\sum_{x=a}^b x = \frac{(a+b)n}{2}.$$

Thus,

$$\mathbb{E}[X] = \frac{1}{n} \cdot \frac{(a+b)n}{2} = \frac{a+b}{2}.$$

So the mean is the midpoint of the interval $[a, b]$.

Discrete Uniform: Variance

$$\mathbb{E}[X^2] = \frac{1}{n} \sum_{x=a}^b x^2.$$

Recall the closed form:

$$\sum_{x=1}^m x^2 = \frac{m(m+1)(2m+1)}{6}.$$

Therefore,

$$\sum_{x=a}^b x^2 = \sum_{x=1}^b x^2 - \sum_{x=1}^{a-1} x^2 = \frac{b(b+1)(2b+1)}{6} - \frac{(a-1)a(2a-1)}{6}.$$

Discrete Uniform: Variance cont.

Now subtract $(\mathbb{E}[X])^2 = \left(\frac{a+b}{2}\right)^2$:

$$V(X) = \frac{1}{n} \left[\frac{b(b+1)(2b+1)}{6} - \frac{(a-1)a(2a-1)}{6} \right] - \left(\frac{a+b}{2} \right)^2.$$

Simplifying this expression yields

$$V(X) = \frac{(b-a+1)^2 - 1}{12}.$$

Discrete Uniform: Typos Example

Typographical errors in books published by a company are uniformly distributed between 6 and 14 per book. Determine the probability mass function, the expected value, the variance.

$X \sim \text{Discrete Uniform on } \{6, 7, \dots, 14\}, \quad n = 14 - 6 + 1 = 9.$

The probability mass function (pmf) is:

$$f(x) = \Pr(X = x) = \begin{cases} \frac{1}{9}, & x \in \{6, 7, \dots, 14\}, \\ 0, & \text{otherwise.} \end{cases}$$

Discrete Uniform: Typos Example cont.

Expected value:

$$\mathbb{E}[X] = \frac{a + b}{2} = \frac{6 + 14}{2} = 10.$$

Variance:

$$\begin{aligned} V(X) &= \frac{(b - a + 1)^2 - 1}{12} \\ &= \frac{(14 - 6 + 1)^2 - 1}{12} \\ &= \frac{9^2 - 1}{12} = \frac{80}{12} = \frac{20}{3}. \end{aligned}$$

Discrete Uniform: Typos Example cont.

What is the probability that a randomly selected book has fewer than 8 typographical errors?

$$\Pr(X < 8) = \Pr(X = 6 \text{ or } X = 7) = \Pr(X = 6) + \Pr(X = 7).$$

Using the pmf $f(x) = \frac{1}{9}$ for $x = 6, \dots, 14$,

$$\Pr(X < 8) = \frac{1}{9} + \frac{1}{9} = \frac{2}{9}.$$

Discrete Uniform: Digits Example

A telephone number is selected at random from a directory.

- ▶ What is the probability that the last digit is exactly 6?

Let X denote the last digit of a randomly selected telephone number. Assume each digit $0, 1, \dots, 9$ is equally likely.

Then $\Pr(X = 6) = \frac{1}{10}$.

- ▶ What is the probability that the last digit is less than 3?

$$\Pr(X < 3) = \Pr(X \in \{0, 1, 2\}) = \frac{3}{10}.$$

Bernoulli Trials & Bernoulli Distribution

A sequence of trials is **Bernoulli** if:

- ▶ each trial has exactly two outcomes (success/failure),
- ▶ the success probability p is constant across trials, and
- ▶ trials are independent.

Let $X \in \{0, 1\}$ encode failure/success with

$$f(x) = \begin{cases} 1 - p, & x = 0, \\ p, & x = 1. \end{cases} \quad \mathbb{E}[X] = p, \quad V(X) = p(1 - p).$$

Examples: tossing a coin, checking if a component is defective or non-defective etc.

Binomial $\text{Bin}(n, p)$

For n independent Bernoulli(p) trials, let X be the number of successes. Then, X follows a **Binomial distribution**, $X \sim \text{Binomial}(n, p)$ with,

$$f(x) = \binom{n}{x} p^x (1-p)^{n-x}, \quad x = 0, 1, \dots, n,$$

$$\mathbb{E}[X] = np, \quad V(X) = np(1-p).$$

Note: $\binom{n}{k} = \frac{n!}{k!(n-k)!}$ read as “ n choose k ” is the number of combinations of n items taken k at a time.

Binomial: Calls Example

Phone lines to a company are busy 40% of the time. Busy lines on successive calls are independent. Suppose 8 calls are placed. What is the probability that exactly 3 calls goes through?

Define “success” = *call goes through*. Let X be the number of calls that go through out of $n = 8$ calls. Each call goes through with probability $p = 1 - 0.40 = 0.60$.

- ▶ The events are independent, so $X \sim \text{Binomial}(n = 8, p = 0.6)$.
- ▶ The probability that exactly 3 calls go through is

$$\Pr(X = 3) = \binom{8}{3} (0.6)^3 (0.4)^5 = 0.124.$$

Binomial: Calls Example cont.

What is the probability that at most 5 calls will go through?

Recall that $X \sim \text{Binomial}(n = 8, p = 0.6)$.

$$\Pr(X \leq 5) = \Pr(X = 0) + \Pr(X = 1) + \dots + \Pr(X = 5)$$

$$\Pr(X \leq 5) = \sum_{x=0}^5 \binom{8}{x} (0.6)^x (0.4)^{8-x} \approx 0.685.$$

Note: This corresponds to the Binomial cumulative distribution function (CDF) $F(k) = \Pr(X \leq k)$.

Binomial: Calls Example cont.

What is the probability that at least 4 calls go through?

We want

$$\Pr(X \geq 4) = 1 - \Pr(X \leq 3).$$

$$F(3) = \Pr(X \leq 3) = \sum_{x=0}^3 \binom{8}{x} (0.6)^x (0.4)^{8-x} = 0.1737.$$

Then,

$$\Pr(X \geq 4) = 1 - 0.1737 \approx 0.8263.$$

Binomial: Calls Example cont.

What is the expected value, variance and standard deviation?

Substitute $n = 8$, $p = 0.6$:

- ▶ $\mathbb{E}[X] = \mu = 8 \times 0.6 = 4.8$. On average, 4.8 out of 8 calls will go through when $p = 0.6$.
- ▶ $V(X) = \sigma^2 = 8 \times 0.6 \times 0.4 = 1.92$,
- ▶ $SD(X) = \sigma = \sqrt{1.92} \approx 1.3856$.

The actual number of calls that go through will typically vary by about 1 to 2 around the mean.

Binomial: Guessing Example

A multiple-choice test contains twenty questions each with four answer choices. A student is guessing on each question.

Let X be the number of correct answers on the test. Does X follow a binomial distribution?

- ▶ Fixed number of trials: $n = 20$ questions.
- ▶ Each trial has two possible outcomes: correct or incorrect.
- ▶ The probability of success (correct answer) is the same on each trial: $p = \frac{1}{4} = 0.25$.
- ▶ The trials are independent: each question is answered without affecting the others.

Binomial: Guessing Example cont.

What is the probability that the student will have more than 10 correct answers?

$X \sim \text{Binomial}(n = 20, p = 0.25)$.

We want

$$\Pr(X > 10) = \sum_{x=11}^{20} \binom{20}{x} (0.25)^x (0.75)^{20-x} = 0.0104.$$

Equivalently, using the complement rule:

$$\Pr(X > 10) = 1 - \Pr(X \leq 10) = 1 - F(10) = 0.0104.$$

The probability is only 1.04%!

Geometric Geom(p)

Let X be the number of trials needed for the first success in i.i.d. Bernoulli(p) trials. Then, we say X follows a **Geometric distribution** with,

$$f(x) = (1 - p)^{x-1}p, \quad x = 1, 2, \dots,$$

$$\mathbb{E}[X] = \frac{1}{p}, \quad V(X) = \frac{1 - p}{p^2}.$$

Note: In contrast to Binomial, there is no fixed number of trials.

Geometric: Gene Example

In a study, volunteers are tested for a gene that is known to increase the risk for a certain disease. The probability that a person carries the gene is 0.1.

What is the probability that exactly four persons need to be tested to detect the first one with the gene?

- ▶ Let X be the trial on which the first person with the gene is detected.
- ▶ Each test is a Bernoulli trial with probability of success $p = 0.1$ (person carries the gene), and the tests are independent.

Geometric: Gene Example cont.

We have $X \sim \text{Geometric}(p = 0.1)$.

The probability mass function of the geometric distribution is

$$\Pr(X = k) = (1 - p)^{k-1}p, \quad k = 1, 2, \dots$$

For $k = 4$,

$$\Pr(X = 4) = (1 - 0.1)^{4-1}(0.1) = (0.9)^3(0.1) = 0.0729.$$

The probability that the first person with the gene is detected on the 4th test is 0.0729.

Geometric: Gene Example cont.

What is the probability that at least five people need to be tested?

$$\Pr(X \geq 5) = 1 - F(4) = 1 - \Pr(X \leq 4).$$

Where,

$$\Pr(X \leq 4) = \sum_{x=1}^4 (0.9)^{x-1} (0.1) = 0.3439.$$

Therefore,

$$\Pr(X \geq 5) = 0.6561.$$

Geometric: Gene Example cont.

Find the expectation, variance and standard deviation

- ▶ $\mathbb{E}[X] = \mu = \frac{1}{0.1} = 10$. On average, one would need to test about 10 people before finding the first person who carries the gene.
- ▶ $Var(X) = \sigma^2 = \frac{1-0.1}{(0.1)^2} = \frac{0.9}{0.01} = 90$,
- ▶ $SD(X) = \sigma = \sqrt{90} \approx 9.49$.

The actual number of tests required can vary quite widely around the mean.

Geometric: Examples

Calls connect with probability 0.2:

- ▶ What is the probability the first call that connects is the sixth call?

$$\mathbb{P}(X = 6) = (0.8)^5(0.2) \approx 0.065.$$

- ▶ What is the probability that it will require at most five calls to connect?

$$\mathbb{P}(X \leq 5) = \sum_{x=1}^5 (0.8)^{x-1}(0.2) \approx 0.672.$$

Negative Binomial $\text{NegBin}(r, p)$

Let X be the number of trials needed to obtain the r -th success in i.i.d. Bernoulli(p) trials. Then, we say X follows a **Negative Binomial distribution** with,

$$f(x) = \binom{x-1}{r-1} (1-p)^{x-r} p^r, \quad x = r, r+1, \dots$$

$$\mathbb{E}[X] = \frac{r}{p}, \quad \text{Var}(X) = \frac{r(1-p)}{p^2}.$$

Negative Binomial: Bit Example

The probability that a bit is transmitted in error is 0.1.
Assuming independent transmissions, what is the probability that the fourth error will occur at the twelfth transmission?

Let X be the trial on which the r -th error occurs.
Then, $X \sim \text{NegBin}(r = 4, p = 0.1)$ and $x = 12$.

$$\Pr(X = 12) = \binom{11}{3} (0.1)^4 (0.9)^8 = 0.00711.$$

Negative Binomial: Bit Example cont.

What's the expected value and variance?

For $r = 4$ and $p = 0.1$:

$$\mathbb{E}[X] = \frac{4}{0.1} = 40,$$

$$V(X) = \frac{4(1 - 0.1)}{(0.1)^2} = \frac{4 \times 0.9}{0.01} = \frac{3.6}{0.01} = 360.$$

On average, it takes 40 transmissions to see 4 errors, with high variability.

Hypergeometric

From a population of size N with K successes and $N - K$ failures, select a sample of size n **without replacement**.

Let X be the number of successes in the sample. X follows a **hypergeometric distribution** with,

$$f(x) = \frac{\binom{K}{x} \binom{N-K}{n-x}}{\binom{N}{n}}, \quad x = 0, 1, \dots, n,$$

$$\mathbb{E}[X] = n \frac{K}{N}, \quad V(X) = n \frac{K}{N} \left(1 - \frac{K}{N}\right) \frac{N-n}{N-1}.$$

Note: In contrast to Binomial, the probability of success changes on every trial.

Hypergeometric: Defective Example

In a batch of 30 components, five are defective. A sample of six components is selected without replacement. What is the probability that there will be two defective items in this sample?

Lot with $N = 30$, $K = 5$ defective; sample $n = 6$.

- ▶ $\mathbb{P}(X = 2) = \frac{\binom{5}{2} \binom{25}{4}}{\binom{30}{6}} \approx 0.213$.
- ▶ $\mathbb{E}[X] = 6 \frac{5}{30} = 1$. On average, we expect about one defective item in a sample of six.
- ▶ $V(X) \approx 0.69$, $\sigma \approx 0.83$. Typical deviation from the mean is less than one item.

Poisson $\text{Pois}(\lambda)$

Named after the French mathematician Simeon Denis Poisson.

Expresses the probability of a given number of events occurring in a fixed interval of time or space if:

- ▶ Events occur regularly with a known constant mean rate,
- ▶ Events occur independently of the time since the last event.

It is customary to denote the constant mean rate by λ .

Poisson cont.

The Poisson distribution can be applied to the number of events in specified intervals such as time, distance, area, or volume.

- ▶ Car accidents in a given unit of time,
- ▶ Traffic flow and ideal gap distance,
- ▶ Number of typing errors on a page,
- ▶ Spread of an endangered animal,
- ▶ Failure of a machine in a given unit of time,
- ▶ Number of patients arriving at a hospital emergency center,
- ▶ Calls coming to a call center in a given unit of time.

Poisson cont.

Let X be the number of occurrences in a fixed unit interval. X follows a **Poisson distribution** with mean rate $\lambda > 0$,

$$f(x) = e^{-\lambda} \frac{\lambda^x}{x!}, \quad x = 0, 1, 2, \dots;$$

$$\mathbb{E}[X] = V(X) = \lambda.$$

Poisson: Electrons Example

Number of electrons that are emitted during a decay process has a Poisson distribution with average of six per second.

What is the probability that there will be exactly four electrons emitted per second?

Let X be the number of electrons emitted per second.

We are told $X \sim \text{Poisson}(\lambda = 6)$.

For $k = 4$,

$$\Pr(X = 4) = \frac{e^{-6}6^4}{4!} = 0.1339.$$

Poisson: Electrons Example cont.

Electrons emitted per second with average $\lambda = 6$.

- ▶ At most three electrons emitted per second?

$$\mathbb{P}(X \leq 3) = \sum_{x=0}^3 e^{-6} \frac{6^x}{x!} = 0.151.$$

- ▶ At least five electrons emitted per second?

$$\mathbb{P}(X \geq 5) = 1 - \sum_{x=0}^4 e^{-6} \frac{6^x}{x!} = 0.715.$$

Poisson: Electrons Example cont.

What's the probability of exactly ten electrons emitted per three seconds?

For a time interval of $t = 3$ seconds, the mean rate is $\lambda_t = \lambda \cdot t = 6 \times 3 = 18$.

Thus,

$$X_3 \sim \text{Poisson}(\lambda = 18).$$

Then,

$$\mathbb{P}(X_3 = 10) = e^{-18} \frac{18^{10}}{10!} = 0.062$$